

Discussion Exercise Sheet 1

Exercise 1.1 Sum of Squares (2 bonus points). → Peergrading Aufgabe für heute

Prove by mathematical induction that for every positive integer n ,

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Recall the notation used here: $\sum_{i=1}^n i^2 = 1^2 + 2^2 + \dots + n^2$.

Wir iterieren über die Variable k

Base Case: Sei $k=1$

$$\sum_{i=1}^1 i^2 = 1 = \frac{1(1+1)(2 \cdot 1+1)}{6} \text{ stimmt, deshalb gilt die Aussage für } k=1$$

Induktions-Hypothese: Wir nehmen an, dass die Aussage für ein beliebiges $k \in \mathbb{N}$ gilt,

$$\text{d.h. } \sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$$

Induktions-Schritt: Wir wollen die Aussage für $k+1$ zeigen, d.h. $\sum_{i=1}^{k+1} i^2 = \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$

$$\sum_{i=1}^{k+1} i^2 = \underbrace{\sum_{i=1}^k i^2}_{\text{IH}} + (k+1)^2$$

$$= \underbrace{\frac{k(k+1)(2k+1)}{6}}_{\text{mit 6 erweitern}} + (k+1)^2$$

$$= \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6} \quad \left\{ \begin{array}{l} \text{auf einen Bruch nehmen} \\ (k+1) \text{ ausklammern} \end{array} \right.$$

$$= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \quad \left\{ \begin{array}{l} \text{ausmultiplizieren} \end{array} \right.$$

$$= \frac{(k+1)(k(2k+1) + 6(k+1))}{6}$$

$$= \frac{(k+1)(2k^2 + k + 6k + 6)}{6}$$

$$= \frac{(k+1)(2k^2 + 7k + 6)}{6} \quad \left\{ \begin{array}{l} \text{faktorisieren} \end{array} \right.$$

$$= \frac{(k+1)(k+2)(2k+3)}{6} \quad \left\{ \begin{array}{l} \text{mit } (k+1) \text{ formulieren} \end{array} \right.$$

$$= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6} \quad \square$$

Consider the following claim:

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} \leq \sqrt{n}.$$

A student provides the following induction proof. Is it correct? If not, explain where the mistake is.

Base case: $n = 1$.

$$\frac{1}{\sqrt{1}} \leq 1, \text{ which is true.} \quad \text{correct}$$

Induction hypothesis: Assume the claim holds for $n = k$, i.e.

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k}. \quad \text{correct}$$

Induction step: Then, starting from the claim we need to prove for $n = k + 1$ and using logical equivalences:

$$\begin{aligned} \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} &\leq \sqrt{k+1} \xLeftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k+1} - \frac{1}{\sqrt{k+1}} \\ &\xLeftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k+1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+1}} \end{aligned} \quad \left. \begin{array}{l} \text{korrekte} \\ \text{Umformungen} \end{array} \right\}$$

$$\xRightarrow{\text{X}} \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \leq \frac{k}{\sqrt{k}} = \sqrt{k}, \quad \text{I.H.}$$

which is true, therefore the claim holds by the principle of mathematical induction.

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \quad \text{✗} \Rightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k}}$$

$$\Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k} \quad \text{I.H.}$$

Induktionsschritt geht in die falsche Richtung: $k+1 \rightarrow k$

Exercise 1.3 Asymptotic growth – part i (2 bonus points).

Recall the concept of asymptotic growth that we introduced in Exercise sheet 0: If $f, g : \mathbb{N}_{\geq 10} \rightarrow \mathbb{R}^+$ are two functions, then:

- We say that f grows asymptotically slower than g if $\lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = 0$. If this is the case, we also say that g grows asymptotically faster than f .

Prove or disprove each of the following statements.

- (a) $f(m) = 100m^3 + 10m^2 + m$ grows asymptotically slower than $g(m) = 0.001 \cdot m^5$.

$$\lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} =$$

- (b) $f(m) = \log(m^3)$ grows asymptotically slower than $g(m) = (\log m)^3$.

$$\lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} =$$

- (c) $f(m) = e^{2m}$ grows asymptotically slower than $g(m) = 2^{3m}$.

Hint: Recall that for all $n, m \in \mathbb{N}$, we have $n^m = e^{m \ln n}$. (very bad hint)

$$\lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} =$$

Exercise 1.4 Asymptotic growth – part ii.

Prove or disprove each of the following statements.

- (a)* If $f(m), g(m) > 1$ and $f(m)$ grows asymptotically slower than $g(m)$, then $\log(f(m))$ grows asymptotically slower than $\log(g(m))$.

(a) Prove the following inequality by mathematical induction

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2n-1}{2n} \leq \frac{1}{\sqrt{3n+1}}, \quad n \geq 1.$$

In your solution, you should address the base case, the induction hypothesis and the induction step.

Base Case:

Induktions-Hypothese:

Induktions-Schritt:

O-Notation

f und g sind Funktionen von N nach $\mathbb{R}^+$

Definition 1 (O-Notation). For $f : N \rightarrow \mathbb{R}^+$, es existiert eine positive Konstante C

$$O(f) := \{g : N \rightarrow \mathbb{R}^+ \mid \exists C > 0 \forall n \in N g(n) \leq C \cdot f(n)\}.$$

Definitionszeichen $\nearrow$ $\{g : N \rightarrow \mathbb{R}^+ \mid$
 $\uparrow$ bedingt dass $\exists C > 0$ sodass für alle g ist höchstens um einen
 Eingaben $n \in N$ konstanten Faktor grösser als f
 gilt dass $\forall n \in N g(n) \leq C \cdot f(n)\}$

- $O(f)$ ist die Menge aller Funktionen g , die asymptotisch höchstens so schnell wie f wachsen
- Was f und g am Anfang machen (für kleine n) macht ist egal
- Wir ignorieren konstante Faktoren
- Beispiel: $f(n) = n$, $g(n) = 3n$, dann gilt $g \in O(f)$ mit $C = 3$
- Wir schreiben oft " $\leq$ " statt " $\in$ "

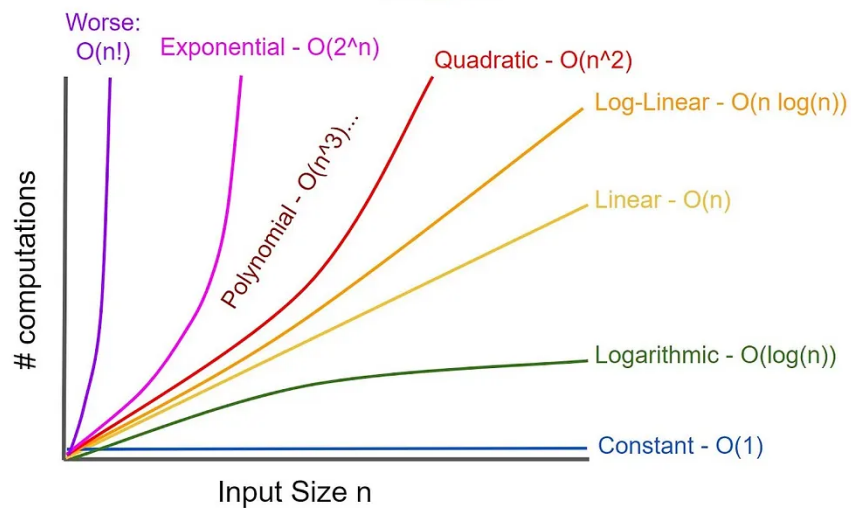
Theorem 1. Let N be an infinite subset of $\mathbb{N}$ and $f : N \rightarrow \mathbb{R}^+$ and $g : N \rightarrow \mathbb{R}^+$.

- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$, then $f \leq O(g)$ and $g \not\leq O(f)$. g wächst schneller
- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = C \in \mathbb{R}^+$, then $f \leq O(g)$ and $g \leq O(f)$. beide wachsen gleich schnell
- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$, then $f \not\leq O(g)$ and $g \leq O(f)$. f wächst schneller

The following theorem can also be helpful when working with O-notation.

Theorem 2. Let $f, g, h : N \rightarrow \mathbb{R}^+$. If $f \leq O(h)$ and $g \leq O(h)$, then

1. For every constant $c > 0$, $c \cdot f \leq O(h)$. "konstante Faktoren ignorieren"
2. $f + g \leq O(h)$. Bsp. $\log n \leq O(n)$ und $2n \leq O(n) \Rightarrow \log n + 2n \leq O(n)$



Aufgaben

Prove or disprove: $3n \leq O(n \log n)$

Prove or disprove: $\log_3(n^3) \leq O(\ln n)$

Prove or disprove: $2^n \leq O(n^3)$

Prove or disprove: $\log_2(n) \cdot \sqrt{n} \leq O(\ln n)$

Prove or disprove: $2^{n \log_2 n} \leq O(2^{n \ln n})$

Order the following functions by their asymptotic growth rate as $n \rightarrow \infty$:

$n!$, $\sqrt{n}$, 2^n , 1 , n^n , $\log n$, n , $n^{\frac{n}{2}}$, n^2

Order the following functions:

n^{10} , $n \log n$, 3^n , $\sqrt{n}$

Order the following functions:

$$n^2 + \log n, \quad n + 100, \quad \sqrt{n} + \log^2 n, \quad n^3 + n$$

Order the following functions:

$$2^n + n^2, \quad n^4 + 3n, \quad n \log n + 5, \quad 3^n + n^3$$

Order the following functions:

$$n^2 + 2^n, \quad n \log n + n^2, \quad n + \sqrt{n}, \quad 10^n + n^3$$

Alte Prüfungsaufgaben

Claim	true	false
$n^3 - n^2\sqrt{n} \leq O(\sqrt{n})$	☐	☐

$2^{(2n)} \leq O(10^n)$	☐	☐
-------------------------	--------------------------	--------------------------

	true	false
$1 \cdot 2 \cdot 3 \cdot \dots \cdot n \leq O(2^n)$	☐	☐

$n^{100} \leq O(2^n)$	☐	☐
-----------------------	--------------------------	--------------------------

$\frac{1}{2}n^2 - \sum_{i=1}^{n-1} i \leq O(n)$	☐	☐
---	--------------------------	--------------------------

$n^{10} \leq O(\log(n)^{100})$	☐	☐
--------------------------------	--------------------------	--------------------------

Induktionsaufgabe

Prove by mathematical induction that the following statement holds for all $n \geq 4$:

$$n! > 2^n$$

Base Case:

Induktions-Hypothese:

Induktions-Schritt: